# Atin Sharma

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## Profile

Full-Stack Product Engineer specializing in zero-to-one development of conversational AI, agentic workflows, and production RAG pipelines. Founding engineer with experience shipping production systems—including a WhatsApp-native AI assistant deployed to ∼40 pilot SMBs—while reducing API latency by 40% and deployment cycle time by 30% through TypeScript-driven architecture and database optimizations.

## Professional Experience

**OpenBiz Software India Pvt Ltd** — Jun 2025 – May 2026
*Founding Engineer* — *Remote / Bangalore*

- VyaparGPT: built full WhatsApp message lifecycle (webhook ingestion → HMAC-SHA256 validation → session context → LLM dispatch → response delivery), serving 40+ SMBs in closed pilot, scaling to 1,000+ active platform users. Multi-turn conversational context via rolling `conversation_history[]`.
- LLM provider-fallback: OpenAI → Google Gemini on HTTP 5xx, 429 rate-limit, or configurable timeout, with response-shape normalisation — zero failure surface exposed to users.
- Gemini Vision API document verification module: extracting structured fields (GST number, business name, invoice amount, dates) from WhatsApp photographs of GST certificates and vendor agreements, eliminating a manual data-entry step in SMB onboarding.
- N+1 fix: rewrote ORM calls to JOIN-based queries, introduced composite indexes on high-frequency WHERE/JOIN columns, and added TTL-based response caching — achieving verified **40% API latency reduction** at 1,000+ user scale.
- Razorpay: HMAC webhook validation, payment-state reconciliation, out-of-order event handling via re-entrant idempotency check — achieving **100% payment-state consistency**.
- CI/CD: GitHub Actions + Vercel, automated testing on every PR, staging deploys on merge, production gated behind explicit approval — **30% reduction in deployment cycle time** and zero broken-build incidents over four months.
- Website Builder product: shipped within 6-week window at zero headcount increase by auditing and reusing ∼60% of shared infrastructure (auth, Supabase layer, Razorpay billing, CI/CD).

**Vellore Institute of Technology** — Jan 2025 – Nov 2025
*Research Contributor* — *Vellore (On-campus)*

- Contributed to research on **Blockchain-Based LLM Model Using Fully Homomorphic Encryption (FHE) for Academic Records** — investigating how FHE enables computation on encrypted student data without decryption, with blockchain providing an immutable audit trail for credential issuance and verification.
- Evaluated architectural trade-offs between on-chain data storage and off-chain encrypted record pointers, assessing gas cost constraints versus data privacy guarantees in an academic credential provenance system.
- Explored how model inference could operate on privacy-preserving representations of sensitive academic data using FHE-compatible cryptographic primitives, without exposing plaintext records.

## Projects

**Nimbus — AI RAG Workspace** | *Next.js 14, TypeScript, PostgreSQL, pgvector, Prisma, NextAuth.js, Groq API, OpenAI*In Develop

- Production-grade RAG pipeline: async queue-based ingestion (status states: pending → processing → indexed → error), overlap chunking to prevent semantic loss at chunk boundaries, embedding generation, and storage in a pgvector extension column within PostgreSQL.
- Cosine similarity search via pgvector (`embedding <=> query_vector`) co-locates vector data with relational metadata (`user_id`, `document_id`) for hybrid SQL + semantic filtering — sub-800ms retrieval target.
- Provider-agnostic LLM abstraction layer: single config flag (`provider: "groq" | "openai" | "anthropic"`) routes generation calls — switching inference providers requires zero code changes.
- Streaming responses via Groq API delivered as SSE, providing real-time token-by-token document Q&A. Ingestion pipeline includes 429 retry logic (exponential backoff) and per-document status tracking.

**DocGPT — RAG Document Assistant** | *React 19, Python (Django), ChromaDB, OpenAI, Docker*

- Full Django RAG pipeline: PDF extraction via PyPDF2 with overlap chunking (∼500–1000 tokens, ∼50–100 overlap), **local embedding** via all-MiniLM-L6-v2 (384-dimensional, offline — zero API cost), ChromaDB cosine retrieval, GPT-4o-mini generation.
- Containerized Django backend + ChromaDB as Docker services; prompt structure: system context + k-nearest retrieved chunks + user query for grounded generation.

**Autonomous Web Agent** | github.com/atinsharma24/auto-agent | *Python, Groq API, BrowserOS (MCP), Claude 3.5 Sonnet*

- Dual-agent architecture: Executor (Claude 3.5 Sonnet) handles complex multi-step DOM + form reasoning; Watcher (Groq fast inference) validates page state between every transition — separating expensive reasoning from high-frequency cheap polling.
- MCP (Model Context Protocol) via BrowserOS gives the Executor structured tool-level access to browser DOM (inspect labels, types, values), form-fill, and navigation — intent-driven traversal with no brittle CSS selectors.

- Persistent local queue manager (status: pending → in_progress → applied / failed / escalated) with URL deduplication; WAF/CAPTCHA detection via DOM fingerprint — Watcher escalates to human, Executor resumes from last checkpoint.

**Automated Content Intelligence Pipeline** | *React 19, TypeScript, Laravel, Node.js, OpenAI, Gemini, Docker, GitHub Actions, Pup*

- Puppeteer headless Chromium (`waitUntil: 'networkidle2'`) for SPA scraping; dual-model AI rewriting (Gemini primary → OpenAI fallback per article); React 19 split-screen Diff View with per-article state machine (pending_review → approved → published).
- Dockerized microservices (Scraper, AI Rewriter, Laravel API, React Frontend) with unless-stopped restart policy; GitHub Actions cron schedule drives fully automated ingestion and review cycles.

## Skills

**Languages:** JavaScript/TypeScript, Python, SQL, Java
**Frontend:** React (v18/19), Next.js, React Native, Redux Toolkit, TailwindCSS
**Backend/DB:** Node.js, Express, NestJS, Django, PostgreSQL, MongoDB, pgvector, ChromaDB
**AI/LLM:** OpenAI, Google Gemini, Groq API, RAG Pipelines, LangChain.js, Sentence Transformers, Prompt Engineering
**DevOps:** Docker, GitHub Actions, Vercel, AWS (S3, DynamoDB)
**Tools:** Git, Postman, Socket.IO, Razorpay SDK, VideoSDK, Puppeteer
**Patterns:** WebSocket (Socket.IO), REST, Webhook Validation (HMAC-SHA256), N+1 Query Optimization, Vector Similarity Search (cosine), Provider-Agnostic LLM Layers, Idempotent Event Processing

## Education

**Vellore Institute of Technology** — Expected Graduation: May 2026
*B.Tech in Computer Science and Engineering* — *Vellore*

- Focus on Full Stack Development and AI Systems.

## Profile Snapshot

**Availability:** Immediate Joiner — 0-day notice period
**Work Mode:** Remote-first; open to Bangalore, Hyderabad, Pune, Delhi NCR, Mumbai
**Expected CTC:** 15–18 LPA (INR) — negotiable for high-ownership roles with equity
**Work Authorization:** Indian Citizen — no sponsorship required
**GitHub:** github.com/atinsharma24
**LinkedIn:** linkedin.com/in/atinsharma24